```r
#GATHER Commannd
View(Titanic)
help("Titanic")
Titanic_N=data.frame(Titanic)

install.packages("tidyr")
library(tidyr)

#classwise count (row repetition to column heading)
#spread: No. of columns will increase
Data1=spread(Titanic_N, Class, Freq)
Data2=spread(Titanic_N, Class, Age)

#classwise count (club rows 4 to 7) i. e column to row
#gather: No. of columns will decrease
Data3=gather(Data1, "Class", "Freq", 4:7)

# unite and separate
#Split and merge columns with unite() and separate()

View(storms)
storms=data.frame(storms)
storms_club=unite(storms, "date", year, month, day, sep = "-")

storms_seperate=separate(storms_club, date, c("year", "month", "day"), sep = "-")

install.packages("dplyr")
library(dplyr)


View(CO2)

#titanic, BOD, CO2
data()

#Recall for all points

data(CO2)
View(CO2)
data.frame(AirPassengers)
# Data Wrangling using tidyr

# Installation of Required Packages
install.packages("tidyr")
library(tidyr)
#Import "marketing_campaign" data set
str(marketing_campaign)
summary(marketing_campaign)

# Import Data set "marketing_campaign" using "Import Dataset"

View(marketing_campaign)
#marketing_campaign <- data.frame(marketing_campaign_1_)
str(marketing_campaign)
summary(marketing_campaign)
```

```r
# Collapsing multiple columns into two columns

## Example 1
Y = gather(marketing_campaign, "Cmp", "n", 21:25)
View(Y)
# Observe the last two columns of new data set Y. Also observe the change in
# total number of columns


## Example 2
X = gather(marketing_campaign, "PurchasesType", "Quantity", 16:19)
View(X)
# Observe the last two columns of new data set Y. Also observe the change in
# total number of columns


# Generates multiple columns from two columns

## Example 1
Y1 = spread(marketing_campaign, Marital_Status, Income)
View(Y1)
# Observe the column numbers 28 - 35 of new data set Y1. Also observe the change
# in total number of columns


## Example 2
Y2 = spread(marketing_campaign, Education, Income)
View(Y1)
# Observe the column numbers 28 - 32 of new data set Y1. Also observe the change
# in total number of columns


# splitting a column by a character string separator
Y3 = separate(marketing_campaign, Dt_Customer, c("day", "month", "year"),
              sep = "-")
View(Y2)
# Observe the column numbers 8 - 10 of new data set Y2. Also observe the change
# in total number of columns


# Uniting multiple columns into a single column
Y3 = unite(Y2, "Dt_Customer", year, month, day, sep = "-")
View(Y3)
# Observe the column number 8 of new data set Y3. Also observe the change
# in total number of columns

#For inbuilt dataset
View(population)
```